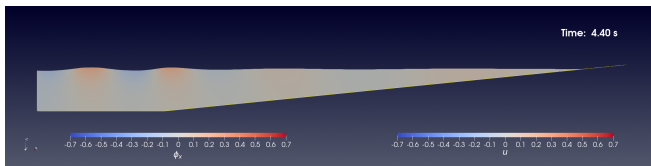
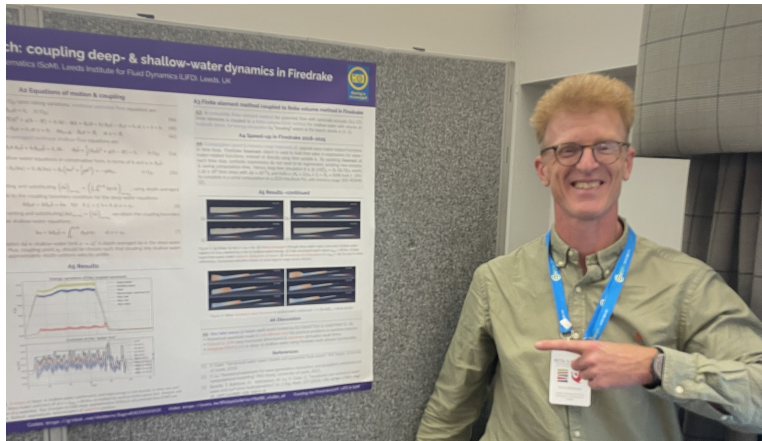


Numerical wavetank with a sloping beach: coupling deep- and shallow-water dynamics in Firedrake

Onno Bokhove, with . . . , Firedrake2026 04-06-2026

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Tributes to Prof Peregrine[†] & Prof Kelmanson

Variational numerical-modelling strategies for the simulation of driven free-surface waves

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Variationally and numerically coupled water-wave and surf zone hydrodynamics

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Abstract

To improve the cost-effectiveness of modelling of wave interaction, a "numerical waverank" is presented whose distinctive novel feature is its ability to couple both deep-water potential-flow and shallow-water models to controllable, prescribed wavermaker motion and beach topography. The coupling is in part obtained via a variational principle approach that guarantees important conservation properties and numerical stability. The model presented is the first fully nonlinear model to couple deep-water (discretised as finite elements) and shallow-water equations (discretised as finite volumes). Resulting simulations of wave generation, propagation and absorption by shallow-water-wave breaking are presented and analysed. A discussion is given on the efficacy of the novel approach.

Keywords Compatible finite-element method · Finite-volume shallow-water method · Potential-flow water-wave dynamics · Shallow-water surf zone · Variationally coupled dynamics

Abstract

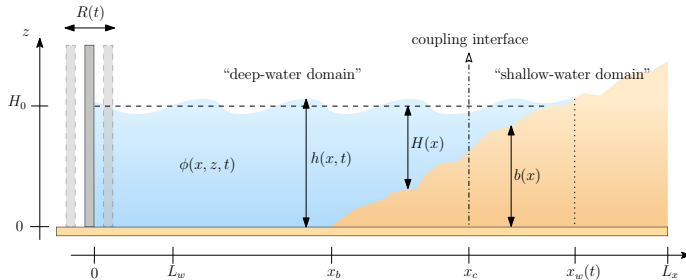
A new tool is developed for simulating three-dimensional (3D) water-wave motion using the first fully variational 3D discretisation in space and time of Luke's variational principle (VP), with additional focus on the formation and analysis of extreme waves generated within in-house experimental wavetanks. This resulting 'numerical wavetank' is able to emulate laboratory sea states in which complex wave-wave and shoaling interactions occur. After first transforming the time-dependent free surface and oscillatory wavemaker into a static rectilinear domain with fixed boundaries using a σ -coordinate transformation, a fully-variational approach is used to derive a system of weak formulations that leads to a non-autonomous space-discrete Hamiltonian system to which robust (stable and mass-conserving) temporal integrators are applied. Specifically, time-discrete VPs with second-order Störmer-Verlet and modified-midpoint time-integration have been derived (and directly implemented through automation) in the finite-element environment *Firedrake* rather than their explicit weak forms, with spectrally-accurate higher-order finite elements. Verification and validation of the new tool are demonstrated via a novel convergence analysis and comparisons of its numerical results with a new analytical solution of three-dimensional two-soliton interaction as well as data post-processed from wavetank experiments.

References

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- 2 Y. Lu. Numerical wavetanks for wave generation, interaction, and dissipation: variational and computational modelling. PhD thesis. University of Leeds, 2025.
- 3 Lu, Y., Gidel, F., Bunnik, T., Bokhove, O., Kelmanson, M. Variational and numerical water- wave and surf-zone hydrodynamics. In: J. Eng. Math. 157 (2026).
<https://doi.org/10.1007/s10665-026-10511-9>
- 4 Lu, Y., Gidel, F., Choi, J., Bokhove, O., Kelmanson, M. Variational numerical-modelling strategies for the simulation of driven free-surface waves. Minor rev. Water Waves.
<https://eartharxiv.org/repository/view/10493/>

Aim

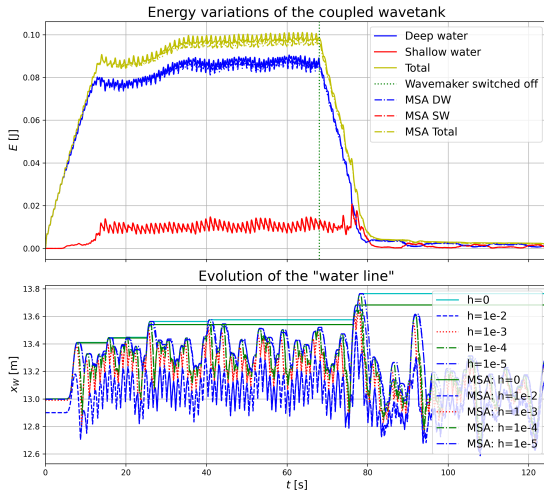
To improve the cost-effectiveness of modelling of wave interactions, a “numerical wavetank” is presented whose distinctive novel feature is its **ability to couple both deep-water potential-flow and shallow-water models** to controllable, prespecified wavemaker motion and beach topography, see Fig. 1:



Achievements

- ▶ **A1:** Model presented is first **fully nonlinear model** to couple deep-water and shallow-water equations, via a variational principle (VP).
- ▶ **A2:** **Coupling** is in part obtained via a VP approach that guarantees important conservation properties and numerical stability.
- ▶ **A3:** Couple deep-water, discretised as **finite element method** in *Firedrake*, and shallow-water equations, discretised as **finite volumes** sec, coupled in *Firedrake*.
- ▶ **A4:** **Speed-up** improvements in *Firedrake*.
- ▶ **A5:** Simulations of **wave generation, propagation & absorption** by shallow-water wave breaking presented.
- ▶ **A6:** Discussion given on efficacy of novel approach & ***Firedrake* improvements**.

A5: Results



A5: Results

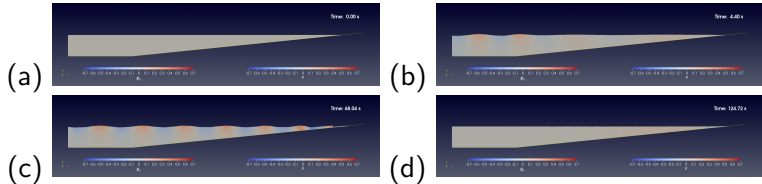


Figure: (a) Water at rest $t = t_0 = 0$ s. (b) **Waves propagate** through deep-water region and enter shallow-water region $t \approx 4.4$ s, marked by a rise in shallow-water energy. (c) Fully-developed water waves $t_{\text{stop}} = 68.04$ s. Energy input from wave-maker **balances dissipation at beach**. (d) **Remaining very long waves** $t = t_{\text{end}} = 124.72$ s due to wave reflections.

A5: Results

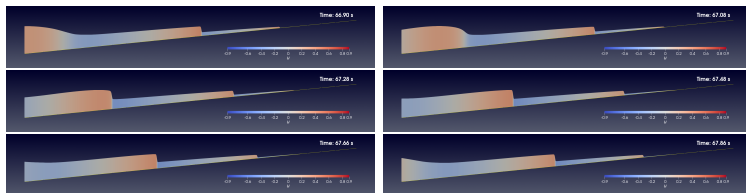
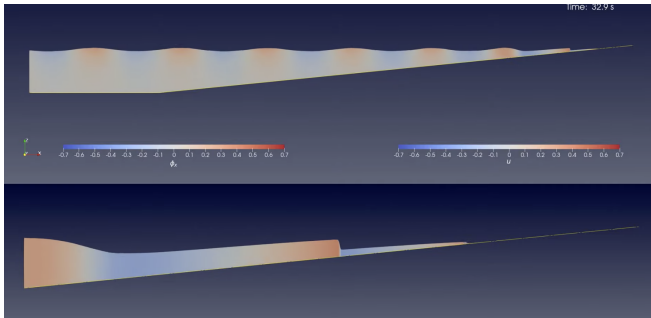


Figure: Wave **shoaling & bore formation** in shallow-water subdomain.
 $t \in [59, 60] T_w = [66.90, 68.04]$ s.

A5: Results



<https://www.youtube.com/watch?v=HFw2ayh2oXk>

A2: Coupling

Coupling-1: Setting and substituting

$$(\delta\check{\phi})_{x=x_c} = \left(\frac{1}{h} \int_b^{b+h} \delta\phi \, dz \right)_{x=x_c}, \text{ using depth-averaged relation}$$

leads to coupling boundary condition for deep-water eqns:

$$h\partial_x\phi = h\partial_x\check{\phi} = hu \quad \text{for} \quad b \leq z \leq b+h \text{ at } x = x_c. \quad (1)$$

Coupling-2: By setting & substituting $(\delta\phi)_{x=x_c} = (\delta\check{\phi})_{x=x_c}$, obtain coupling boundary condition for SWE:

$$hu = h\partial_x\check{\phi} = \int_b^{b+h} \partial_x\phi \, dz \quad \text{at } x = x_c. \quad (2)$$

Rationale: Variation $\delta\check{\phi}$ in SW limit $x \rightarrow x_c^+$ is depth-averaged $\delta\phi$ in DW limit $x \rightarrow x_c^-$. Coupling point x_c should be chosen such that shoaling into SW leads must be approximately depth-uniform velocity profile.

A4: Firedrake & such

- ▶ All (key) Figures reproducible. Full codes on GitHub:
<https://github.com/obokhove/EagreEUEID20202023/tree/master/coupledbeachwave>
- ▶ **A4**: [Computation speed & memory usage improved](#), cf. upgrade wave-maker-related functions in time loop. Firedrake Constant object is used to hold time value in expressions for wave-maker-related functions, instead of directly using time variable t .
- ▶ OB macbook 2020 docker: 2h15min21.46s; Memory usage less than 406MB.
YL macbook native 1:27hr; Memory usage less than 300MB.
M2 macbook docker: 59:38min; memory usage 459.84MB.
M2 macbook native: 39:23min; memory usage: 387.98MB.

A6: Discussion

Key take-aways & future work (funding EU GA607596 & GA859983):

- ▶ Numerical wavetank made is *cost-effective tool* for practical problems in maritime industry.
- ▶ *Extend to 2DH* (two horizontal dimensions) & *automate* derivation weak forms.
- ▶ *Integrate FEM & DG* in deep- & shallow-water using *Firedrake* multi-domain functionality.

A6: (Un)Desirable aspects & future work

- ▶ Technical derivations unwieldy (see JEM2026-paper). Clean, concise derivation (Bokhove 2022)?
- ▶ Weak forms should be automated, done in WW-paper but not in JEM2026-paper.
- ▶ Speed improvements possible via solver optimisation.
- ▶ Higher-order time integrators.
- ▶ Both variational deep- & DG shallow-water models in two coupled (Firedrake) domains (A. & L.).
- ▶ Get rid of coupling point x_c : ask Colin Cotter which model one could use! CC & OB JEM2010.